Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

SEAICE EXTENT INDICATOR

ARCTIC_OMI_SI_extent

ANTARCTIC_OMI_SI_extent

Issue: 2.1

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Approval date by the CMEMS product quality coordination team: November 2025

I CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	25.01.2018	all	First draft version extracted from CMEMS-TARGETED-OSI-QUID-002-V1.2	C. Wettre	T. Lavergne
2.0	12.04.2019	all	Revision for updated OMI that now covers Arctic and Antarctic, and is based on OSI SAF OSI-450 and OSI-430-b	T. Lavergne	C. Wettre
2.1	25.08.2023	All	New version of the OMI, based on the OSI SAF Sea Ice Index (OSI-420) v2.2, itself derived from SIC CDR v3 (OSI-430-a and OSI-430-a)	T. Lavergne	C.Wettre

Product name: OMI_SI_extent

Dataset names: arctic_omi_si_extent, antarctic_omi_si_extent

II INTRODUCTION

This document provides to CMEMS users information on the quality of the OMI monthly Sea Ice Extent (SIE) index for the Arctic and Antarctic.

The data are monthly averaged SIE values. They are extracted and reformatted from the OSI SAF Sea Ice Index (SII, OSI-420) v2.2 product. See the PUM for a description of the CMEMS OMI product and its relation to the OSI SAF SII.

SIE is an integrated quantity and no ground-truth is available for its validation. It is here compared to a similar indicator from the US National Snow and Ice Data Center (NSIDC): the NSIDC Sea Ice Index (SII) v3.

III SCIENTIFIC VALIDATION

An evaluation of the monthly SIE values is documented in dedicated OSI SAF reports for the OSI SAF Sea Ice Index v2.2, accessible from https://osi-saf.eumetsat.int/products/osi-420

In summary, the SIE values from the OSI SAF are similar but not identical to those of NSIDC (stemming from differences in the processing of SIC maps). The OSI SAF values are in general larger than those of NSIDC, with differences reaching 0.5 millions km^2 in Northern Hemisphere summer. Average differences in the Southern Hemisphere are about 0.3 millions km^2 in all months. The two time series agree in terms of SIE trends in both hemispheres, with the NSIDC data showing slightly larger relative trends than OSI SAF during summer, mostly due to the early period in the time series (1979-1987).

The current OSI SAF sea ice index, version 2.3, is still in the demonstration stage, and preliminary quality assessments are available in the Product User Manual (PUM, version 1.3, https://osi-saf.eumetsat.int/products/osi-420). The official index product, version 3.0, is scheduled for review soon, with the final versions of the validation report, PUM, and the Algorithm Theoretical Baseline Document (ATBD) expected to be released in January 2026. Methodological details and complete validation results will be documented in these upcoming OSI SAF publications.

IV QUALITY CHANGES SINCE PREVIOUS VERSION

This is the third version for the OMI SIE.

The second version was based on the OSI SAF Sea Ice Index v2.1, that was itself based on the OSI SAF SIC CDR v2 (OSI-450 and OSI-430-b).

The first version of the OMI had only coverage of the Northern Hemisphere, and was using an older version of the EUMETSAT OSI SAF sea-ice concentration data (OSI-409 and OSI-430).

The new version of the OMI delivered in CMEMS is of higher quality (in accuracy, coverage, and robustness) than the previous versions. Compared to the second version, the new version is closer to the NSIDC values in the Northern Hemisphere.

V REFERENCES:

NSIDC Sea Ice Index v3: http://nsidc.org/data/seaice_index/

Lavergne, T., Sørensen, A. M., Kern, S., Tonboe, R., Notz, D., Aaboe, S., Bell, L., Dybkjær, G., Eastwood, S., Gabarro, C., Heygster, G., Killie, M. A., Brandt Kreiner, M., Lavelle, J., Saldo, R., Sandven, S., and Pedersen, L. T.: Version 2 of the EUMETSAT OSI SAF and ESA CCI sea-ice concentration climate data records, The Cryosphere, 13, 49-78, https://doi.org/10.5194/tc-13-49-2019, 2019.